THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 9: **Symmetry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of letters among H, O, X, I, E with BOTH a vertical and a horizontal line of symmetry is: 1

(a) 5 **(b)** 6 **(c)** 3 **(d)** 4

2. The sum of the angles of rotation of a square and a regular pentagon (in degrees) is: 1

(a) 162 **(b)** 172 **(c)** 152 **(d)** 192

3. A figure has rotational symmetry of order 6 and one of its lines of symmetry. The total number of lines of symmetry it must have is: 1

(a) 8 **(b)** 5 **(c)** 7 **(d)** 6

4. The reflection of a point (x, y) = (6, 1) in the y-axis (vertical line) has x-coordinate: 1

(a) -6 **(b)** 6 **(c)** -5 **(d)** -7

5. A regular polygon is rotated again and again through its smallest angle 24°. The number of turns needed to complete a full turn is: 1

(a) 16 **(b)** 17 **(c)** 14 **(d)** 15

6. A regular 9-gon has side 4 cm. Its perimeter (cm) is: 1

(a) 32 **(b)** 40 **(c)** 36 **(d)** 45

7. The image of a point 12 units from a mirror line is how far from the original point (units)? 1

(a) 36 **(b)** 25 **(c)** 24 **(d)** 12

8. A regular polygon has an angle of rotation 72° and its perimeter is 45 cm. The side length is (cm): 1

(a) 11 **(b)** 9 **(c)** 8 **(d)** 10

9. After a rotation of 108° a regular 10-gon coincides with itself. This rotation is 3 times the smallest rotation. The smallest rotation (degrees) is: 1

(a) 26 **(b)** 66 **(c)** 46 **(d)** 36

10. The number of letters among W, X, Y, Z, H with BOTH a vertical and a horizontal line of symmetry is: 1

(a) 3 **(b)** 4 **(c)** 2 **(d)** 1

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Reflect the point (6, 1) in 2

(a) the y-axis,

(b) the x-axis and give the coordinates.

12. A regular polygon has rotational symmetry with the smallest angle of 20° and each side 3 cm. Find the number of sides and its perimeter. 2

13. A figure rotates and fits after 72° and 216° turns but not at smaller turns. Find the order of symmetry from the smallest angle 72. 2

14. Reflect the point (3, 7) in 2
(a) the y-axis,
(b) the x-axis and give the coordinates.

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A designer wants a symmetrical logo with 10 identical petals. Find the angle between neighbouring petals, the number of lines of symmetry and the total of angles around the centre if 10 petals are placed. 3

16. Triangle ABC has A(2, 2), B(5, 2), C(3, 6) and is reflected in the vertical line x = 8. Find the images of A and B (the reflection of x is $2m - x$). 3

17. A regular 12-gon is rotated about its centre by 210° anticlockwise and the figure looks unchanged. Verify this using the order of symmetry and find the number of rotations of the smallest angle that make 210°. 3

OR

A designer wants a symmetrical logo with 12 identical petals. Find the angle between neighbouring petals, the number of lines of symmetry and the total of angles around the centre if 12 petals are placed. 3

18. Triangle ABC has A(4, 1), B(7, 1), C(5, 5) and is reflected in the vertical line x = 9. Find the images of A and B (the reflection of x is $2m - x$). 3

OR

A regular 5-gon is rotated about its centre by 288° anticlockwise and the figure looks unchanged. Verify this using the order of symmetry and find the number of rotations of the smallest angle that make 288°. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. For a regular 12-gon of side 2 cm: 5
(a) find the number of lines of symmetry,
(b) the angle of rotation,
(c) the perimeter,
(d) the angle turned by a vertex when the polygon is rotated 7 times through the smallest angle.

OR

Two regular polygons have 4 and 5 sides. Find their smallest angles of rotation, the difference, and the least angle (in degrees) that is a turn for both (the LCM of the two angles). 5

20. For a regular 5-gon of side 8 cm: 5
(a) find the number of lines of symmetry,
(b) the angle of rotation,
(c) the perimeter,
(d) the angle turned by a vertex when the polygon is rotated 4 times through the smallest angle.

OR

Two regular polygons have 3 and 5 sides. Find their smallest angles of rotation, the difference, and the least angle (in degrees) that is a turn for both (the LCM of the two angles). 5

END OF QUESTION PAPER — Total: 40 Marks